SOC THREAT HUNTING & INVESTIGATION

TECHNICAL INVESTIGATION REPORT

PhantomStealer Malware — Multi-Layer Network Analysis

DNS Beaconing · HTTP C2 Communication · TLS Encrypted Channel Detection

Analyst	Rohit Aswal
Certification	Certified Ethical Hacker (CEH)
Investigation Period	January – February 2026
Report Date	February 2026
Malware Family	PhantomStealer
Dataset Source	Malware Traffic Analysis (malware-traffic-analysis.net)
PCAP File	2026-01-30-PhantomStealer-infection.pcap
Tools Used	Splunk Enterprise · Wireshark · Kali Linux
Total Events Analysed	239 (DNS: 10 · HTTP: 6 · TLS: 223)
Infected Host	10.1.30.101
Framework	MITRE ATT&CK
Classification	CONFIDENTIAL – Academic Portfolio

Rohit Aswal

SOC Analyst · Threat Hunter · Certified Ethical Hacker (CEH)

LinkedIn: linkedin.com/in/rohit-aswal08 · GitHub: github.com/rohitaswal2108-Roh

> **Disclaimer:** All investigations conducted in a controlled isolated Kali Linux VM using publicly available malware datasets from malware-traffic-analysis.net. No real production systems were involved.

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1. Introduction

1.1 Background and Purpose

This report documents a comprehensive multi-layer network security investigation conducted against a real-world PhantomStealer malware infection. The investigation demonstrates practical SOC analyst threat hunting skills — covering detection, correlation, and attack-chain reconstruction — using industry-standard tooling.

PhantomStealer is an information-stealing malware capable of harvesting browser credentials, session tokens, cryptocurrency wallet data, and Discord tokens from compromised systems. A publicly available packet capture (PCAP) from a confirmed PhantomStealer infection was sourced from malware-traffic-analysis.net, and the complete attack chain was reconstructed across three network protocol layers: DNS, HTTP, and TLS.

The investigation is structured as three interconnected projects. Findings across all three converge to confirm the same Command and Control (C2) infrastructure operating simultaneously over both unencrypted and encrypted channels — a hallmark of modern malware design built for redundancy and stealth.

1.2 Scope of Investigation

The following scope boundaries applied throughout this investigation:

- Dataset: 2026-01-30-PhantomStealer-infection.pcap — captured 30 January 2026
- Protocol Coverage: DNS (Project 1), HTTP (Project 2), TLS/SSL (Project 3)
- Total Events: 239 network events analysed (10 DNS, 6 HTTP, 223 TLS)
- Victim Asset: Internal host 10.1.30.101 on segment 10.1.30.0/24
- Analysis Period: January to February 2026
- Environment: Isolated Kali Linux VM — no connection to production networks

1.3 Tools and Technologies

Tool / Resource	Category	Application in This Investigation
Splunk Enterprise	SIEM Platform	Log ingestion, SPL query development, field extraction, correlation and alerting
Wireshark	Packet Analyser	PCAP handling, protocol filters (dns / http / tls||ssl), CSV export
Kali Linux	Analysis OS	Isolated VM environment for safe malware PCAP handling
MITRE ATT&CK	Threat Framework	Standardised mapping of behaviours to adversary techniques
Malware Traffic Analysis	Dataset Source	Public repository — PhantomStealer infection PCAP used

Table 1: Tools and Technologies Used in This Investigation

1.4 Dataset and Safety Notes

Source: Malware Traffic Analysis — https://www.malware-traffic-analysis.net/

Archive Password: infected_20260130

Malware Family: PhantomStealer — credential stealer targeting browsers, crypto wallets, and Discord tokens

Safety Procedure: PCAP extracted and analysed exclusively inside an isolated Kali Linux virtual machine. The host machine was never exposed to malware traffic or live C2 IPs.

2. Investigation Methodology

2.1 Five-Phase Workflow

Each project followed a consistent five-phase methodology to ensure rigorous, repeatable, and defensible analysis:

1. PCAP Acquisition — Download and safely extract the malware PCAP in an isolated VM
2. Traffic Extraction — Apply Wireshark protocol filter to isolate the target layer (dns / http / tls||ssl)
3. Splunk Ingestion — Export filtered traffic as CSV and ingest into a dedicated Splunk index
4. Threat Hunting — Execute SPL queries for pattern analysis, IOC extraction, and anomaly detection
5. Correlation and Reporting — Cross-reference findings across all protocol layers; map to MITRE ATT&CK

2.2 Splunk Index Architecture

Index Name	Protocol	Events	Purpose
phantom_dns	DNS	10	Domain query frequency and beaconing detection
phantom_http	HTTP	6	C2 session analysis and payload detection
phantom_tls	TLS/SSL	223	Encrypted channel detection via SNI extraction

Table 2: Splunk Index Architecture — Three Dedicated Indexes for Protocol Isolation

2.3 SPL Query Development Approach

Queries were developed iteratively — beginning with broad data exploration, then progressively narrowing toward specific IOC extraction. A key challenge in Project 1 was that Wireshark's CSV export produced a single quoted string per row, requiring manual regex-based field extraction before any meaningful analysis could proceed:

```
| rex field=  raw
  "(?<No>[^"]+)","(?<Time>[^"]+)","(?<Source>[^"]+)",
  "(?<Destination>[^"]+)","(?<Protocol>[^"]+)",
  "(?<Length>[^"]+)","(?<Info>[^"]+)"
```

3. Project 1 — DNS Traffic Analysis

Title: *PhantomStealer DNS Beaconing Detection using Splunk*

MITRE Techniques: T1016 · T1071.004

3.1 Objective

DNS is a foundational layer of C2 communication — even when all subsequent traffic is encrypted, malware must first resolve its C2 domain via DNS. By identifying anomalous DNS query patterns, it is possible to detect beaconing activity and identify the compromised host without decrypting any traffic. This phase analyses the DNS layer of the PhantomStealer infection to extract these signals.

3.2 Data Acquisition and Ingestion

Step 1 — PCAP Download and Extraction

The PhantomStealer PCAP was downloaded from malware-traffic-analysis.net and extracted inside an isolated Kali Linux VM. The archive password follows the site's standard format: infected_20260130.

> **Safety Note:** Malware PCAPs are safe to open (they are network captures, not executable malware) but contain live C2 IP addresses. All handling was performed inside an isolated VM to prevent accidental network contact with C2 infrastructure.

Step 2 — Wireshark DNS Filter and Export

A display filter was applied in Wireshark to isolate DNS traffic:

```
dns
```

Exported via: File → Export Packet Dissections → As CSV → dns_logs.csv. Transferred to Windows Splunk host.

Step 3 — Splunk Index Creation and Ingestion

```
Settings → Indexes → New Index → Name: phantom_dns
Settings → Add Data → Upload → dns_logs.csv
Source Type: csv | Index: phantom_dns
```

> **Result:** 10 DNS events successfully indexed in phantom_dns.

Step 4 — Field Extraction

Splunk treated each CSV row as a raw string. Manual regex extraction was applied:

```
index=phantom_dns
| rex field=_raw "(?<No>[^"]+)","(?<Time>[^"]+)","(?<Source>[^"]+)",
  "(?<Destination>[^"]+)","(?<Protocol>[^"]+)",
  "(?<Length>[^"]+)","(?<Info>[^"]+)"
| table No Time Source Destination Protocol Length Info
```

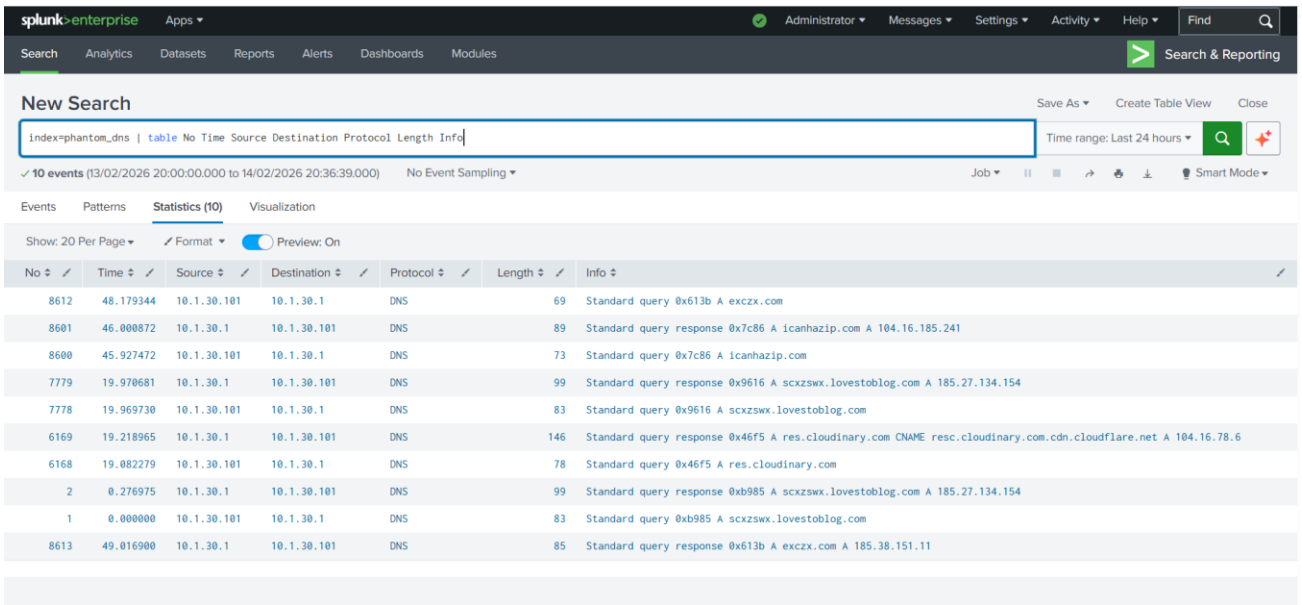


Figure 1: Splunk field extraction result — structured DNS fields visible in table view

3.3 Threat Hunting Analysis

Step 5 — Domain Frequency Analysis

The primary beaconing indicator is repeated DNS queries to the same external domain. This SPL query extracts domain names from the Info field and ranks them by frequency:

```
index=phantom_dns
| rex field=Info "A (?<domain>[a-zA-Z0-9\.-]+\.[a-zA-Z]{2,})"
| stats count by domain
| sort -count
```

Domain Queried	Count	Analyst Assessment
scxzswx.lovestoblog.com	4	CRITICAL — Beaconing confirmed. Random subdomain on free hosting. Primary C2 domain.
exczx.com	2	SUSPICIOUS — Short generated domain. Confirmed secondary C2 in TLS analysis.
icanhazip.com	2	SUSPICIOUS USAGE — Legitimate IP-lookup service abused for public IP reconnaissance.
res.cloudinary.com	2	LEGITIMATE — CDN service. Also appears in TLS; further correlation recommended.

Table 3: DNS Domain Frequency Analysis — All 10 Events

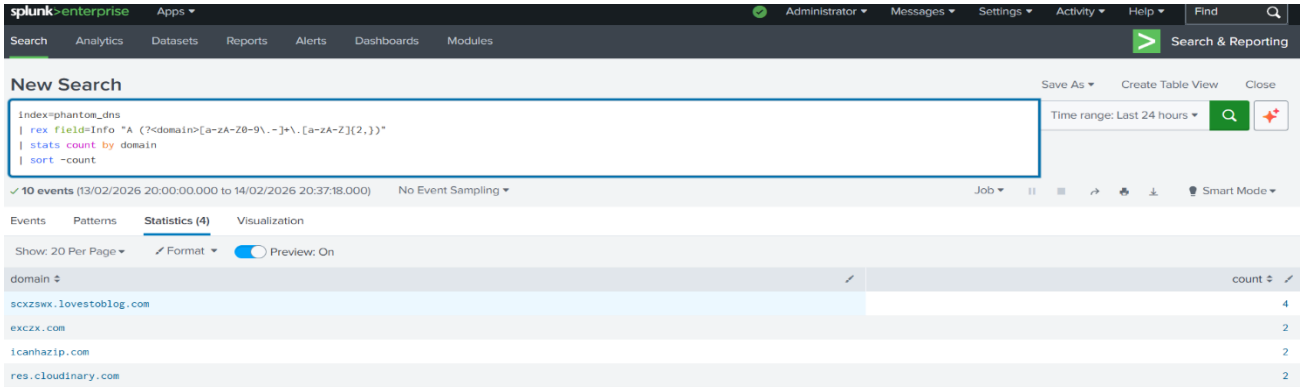


Figure 2: Domain frequency analysis — scxzswx.lovestoblog.com leads with 4 queries confirming beaconing

The domain scxzswx.lovestoblog.com exhibits three concurrent indicators of malicious C2 use: (1) the subdomain name is random in appearance, consistent with DGA or temporary callback subdomains; (2) it is hosted on a free blogging platform — a disposable infrastructure tactic; and (3) it has the highest query frequency at four queries, which is the defining characteristic of beaconing behaviour.

Step 6 — Beaconing Detection

This query filters specifically for domains queried more than once — the minimum condition for beaconing:

```
index=phantom_dns
| rex field=Info "A (?<domain>[a-zA-Z0-9\.-]+\.[a-zA-Z]{2,})"
| stats count by domain
| where count > 1
```

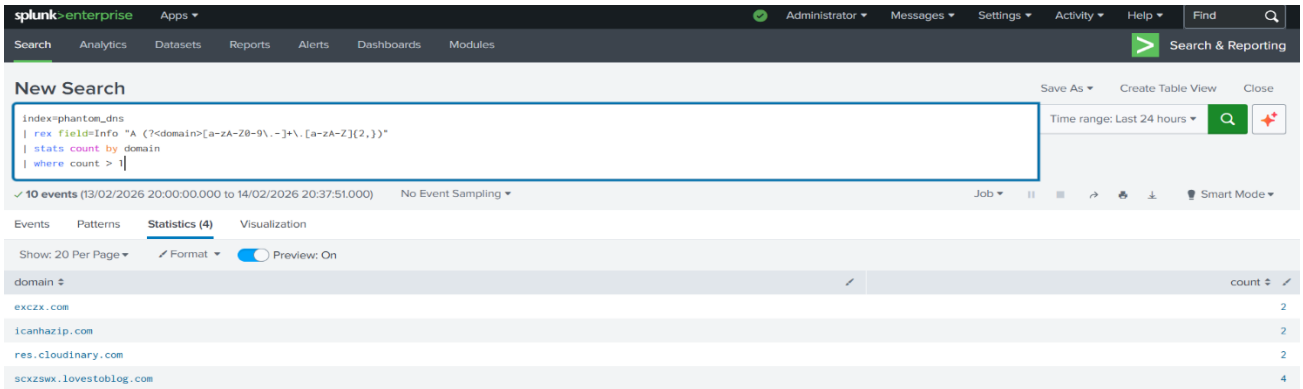


Figure 3: Beaconing detection — all domains with query count greater than 1; scxzswx.lovestoblog.com confirmed

> **Key Finding:** scxzswx.lovestoblog.com with 4 DNS queries is confirmed as an active C2 beaconing domain. This repeated callback pattern is consistent with PhantomStealer's heartbeat mechanism.

Step 7 — Compromised Host Identification

This query identifies which internal endpoint initiated the suspicious DNS queries:

```
index=phantom_dns "Standard query"
| rex field=Info "A (?<domain>[a-zA-Z0-9\.-]+\.[a-zA-Z]{2,})"
| stats count by Source domain
| sort -count
```

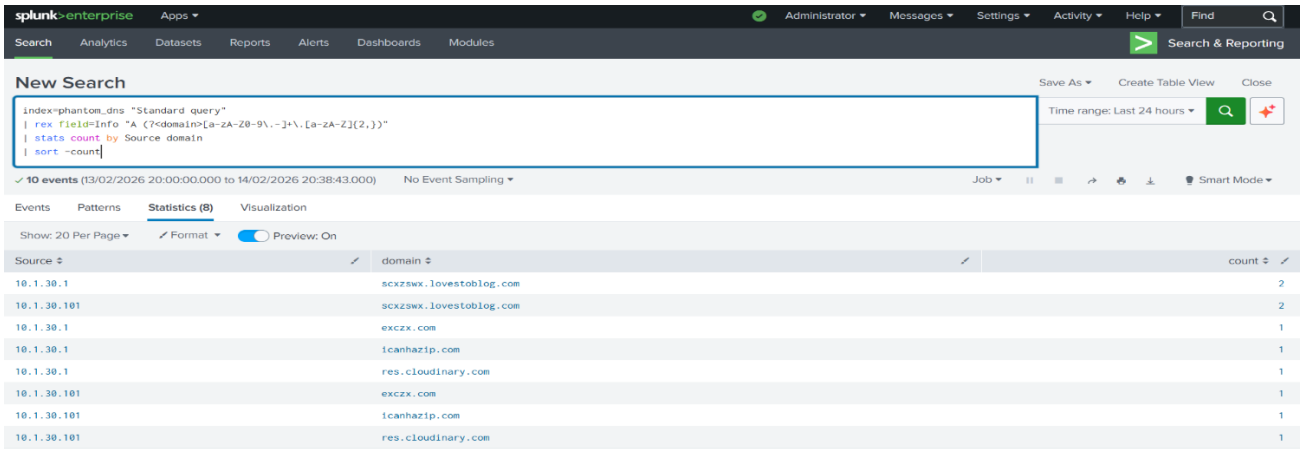


Figure 4: Infected host identification — 10.1.30.101 confirmed as sole source of all malicious DNS queries

> **Compromised Host Confirmed:** Internal IP 10.1.30.101 generated all suspicious DNS queries to the identified C2 domains.

3.4 Project 1 — Key Findings and IOCs

Type	Value	Notes
C2 Domain	scxzswx.lovestoblog.com	Primary C2 — 4 queries, beaconing confirmed
C2 Domain	exczx.com	Secondary C2 — 2 queries, also in TLS analysis
C2 IP	185.27.134.154	Resolved IP for scxzswx.lovestoblog.com
C2 IP	185.38.151.11	Resolved IP for exczx.com
Infected Host	10.1.30.101	Sole originator of all malicious DNS queries
Recon Domain	icanhazip.com	Used for public IP discovery before C2 contact

Table 4: Project 1 Indicators of Compromise

Attack timeline — DNS phase:

```
T+0s   Host 10.1.30.101 compromised — PhantomStealer payload executed
T+1s   DNS: icanhazip.com queried → public IP discovered (reconnaissance)
T+2s   DNS query #1: scxzswx.lovestoblog.com → C2 resolution
T+3s   DNS query #1: exczx.com → secondary C2 resolution
T+4s   DNS queries #2,#3,#4: scxzswx.lovestoblog.com → beaconing confirmed
T+5s   C2 established → proceeds to HTTP communication (Project 2)
```

3.5 MITRE ATT&CK Mapping — Project 1

ID	Technique	Tactic	Evidence
T1016	System Network Configuration Discovery	Discovery	icanhazip.com queried for public IP before C2 contact
T1071.004	App Layer Protocol: DNS	Command & Control	4 repeated queries to scxzswx.lovestoblog.com — beaconing

Table 5: MITRE ATT&CK Mapping — Project 1

4. Project 2 — HTTP Traffic Analysis

Title: *PhantomStealer HTTP C2 Communication Investigation*

MITRE Techniques: T1016 · T1071.001 · T1105 · T1102

4.1 Objective

Project 2 builds directly on DNS findings. Having identified the C2 domain and resolved IP through DNS, this phase analyses the HTTP layer to understand what the malware did once it established contact with the C2 server — specifically identifying downloaded payloads and reconstructing the attack timeline.

> **Cross-Protocol Link:** C2 IP 185.27.134.154 was first identified in Project 1 as the DNS resolution for scxzswx.lovestoblog.com. Confirming HTTP traffic to this IP validates that DNS beaconing led directly to active C2 communication.

4.2 Data Acquisition and Ingestion

The same PCAP was re-opened in Wireshark with an HTTP display filter:

```
http
```

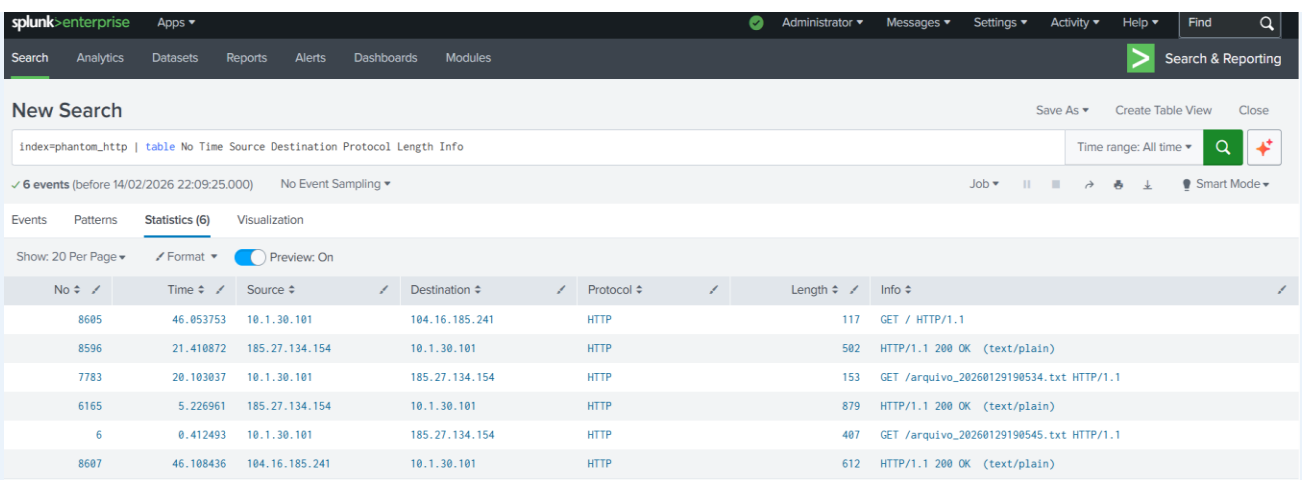
6 HTTP transactions identified. Exported as http_logs.csv and ingested into phantom_http.

> **Result:** 6 HTTP events successfully indexed in phantom_http.

4.3 Threat Hunting Analysis

Step 3 — Traffic Overview

```
index=phantom_http | table Time Source Destination Info
```



No	Time	Source	Destination	Protocol	Length	Info
8605	46.053753	10.1.30.101	104.16.185.241	HTTP	117	GET / HTTP/1.1
8596	21.410872	185.27.134.154	10.1.30.101	HTTP	502	HTTP/1.1 200 OK (text/plain)
7783	20.103037	10.1.30.101	185.27.134.154	HTTP	153	GET /arquivo_20260129190534.txt HTTP/1.1
6165	5.226961	185.27.134.154	10.1.30.101	HTTP	879	HTTP/1.1 200 OK (text/plain)
6	0.412493	10.1.30.101	185.27.134.154	HTTP	407	GET /arquivo_20260129190545.txt HTTP/1.1
8607	46.108436	104.16.185.241	10.1.30.101	HTTP	612	HTTP/1.1 200 OK (text/plain)

Figure 5: HTTP traffic overview — all 6 transactions from 10.1.30.101 to two external IPs

Step 4 — Source and Destination Analysis

```
index=phantom_http | stats count by Source Destination
```

Source IP	Destination IP	Sessions	Assessment
10.1.30.101	**185.27.134.154**	4	C2 Communication — confirmed C2 IP from DNS resolution
10.1.30.101	**104.16.185.241**	2	Reconnaissance — icanhazip.com for public IP discovery

Table 6: HTTP Source-Destination Session Analysis

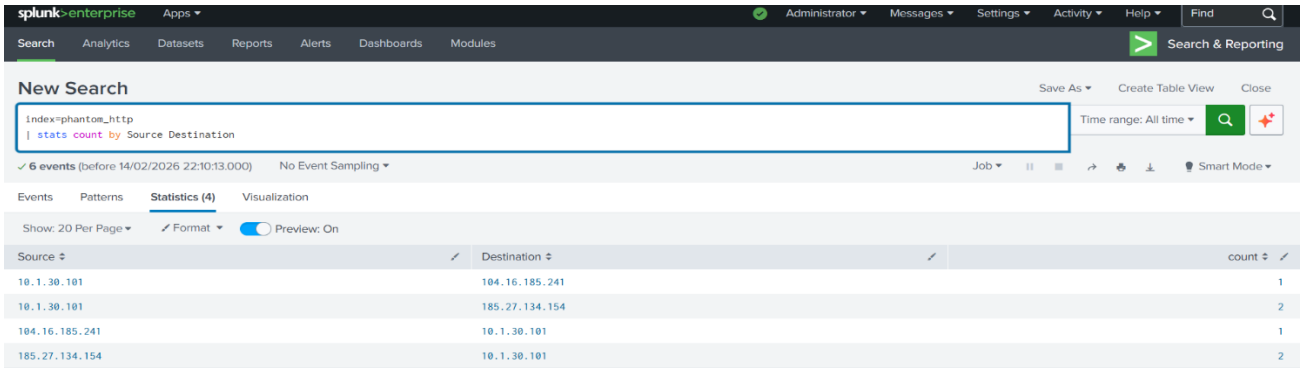


Figure 6: Source-destination analysis — 4 sessions to C2 IP, 2 sessions to icanhazip.com

Step 5 — Malware Config File Detection

This query extracts file paths from HTTP GET requests to identify downloaded payloads:

```
index=phantom_http
| rex field=Info "GET (?<filename>/[^\s]+)"
| where isnotnull(filename)
| table Time Source Destination filename
```

File Path	Timestamp	Server IP	HTTP Status	Significance
/arquivo_20260129190534.txt	2026-01-29 19:05:34	185.27.134.154	**200 OK**	Config file #1 — timestamped automated generation
/arquivo_20260129190545.txt	2026-01-29 19:05:45	185.27.134.154	**200 OK**	Config file #2 — updated config, 11s after first

Table 7: Downloaded Configuration Files Detected via HTTP GET

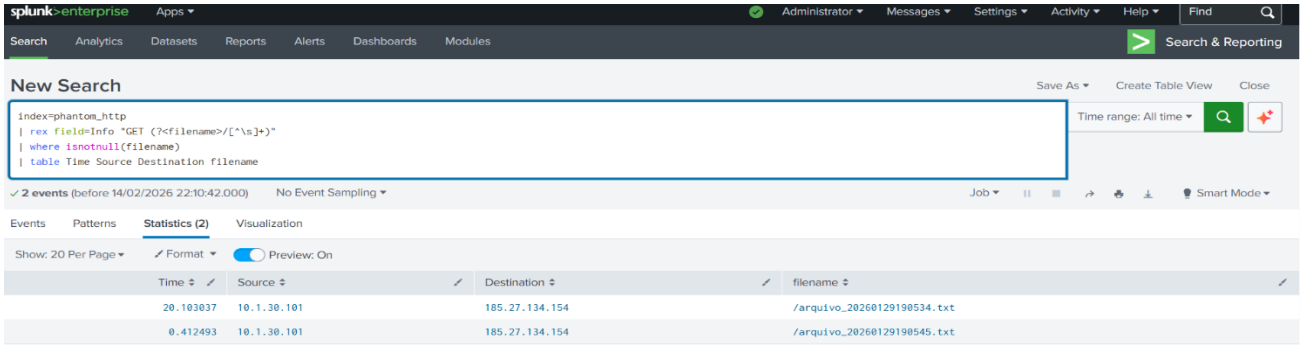


Figure 7: Suspicious file extraction — timestamped arquivo_*.txt config files confirmed downloaded from C2

Two configuration files were downloaded from the C2 server. The name "arquivo" is the Portuguese word for "file" — a strong indicator of a Portuguese-speaking threat actor, consistent with known Brazilian cybercrime groups operating PhantomStealer. The numeric timestamps embedded in filenames (20260129190534 and 20260129190545) suggest automated per-victim config generation on the C2 server. The 11-second gap indicates a two-stage configuration update process.

Step 6 — HTTP Method Distribution

```
index=phantom_http
| rex field=Info "^(?<http_method>GET|POST|HTTP)"
| stats count by http_method
```

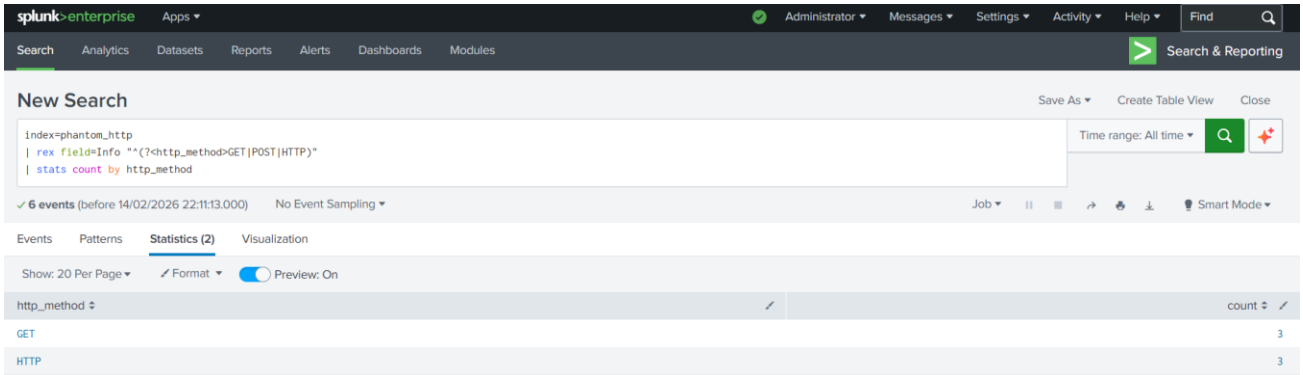


Figure 8: HTTP method distribution — only GET requests observed; all returned HTTP 200 OK

Only GET requests were observed — the malware was purely downloading configuration. The absence of POST requests indicates data exfiltration either was not captured in this window or occurs over the encrypted TLS channel analysed in Project 3.

Step 7 — Attack Timeline

```
index=phantom_http | timechart count
```

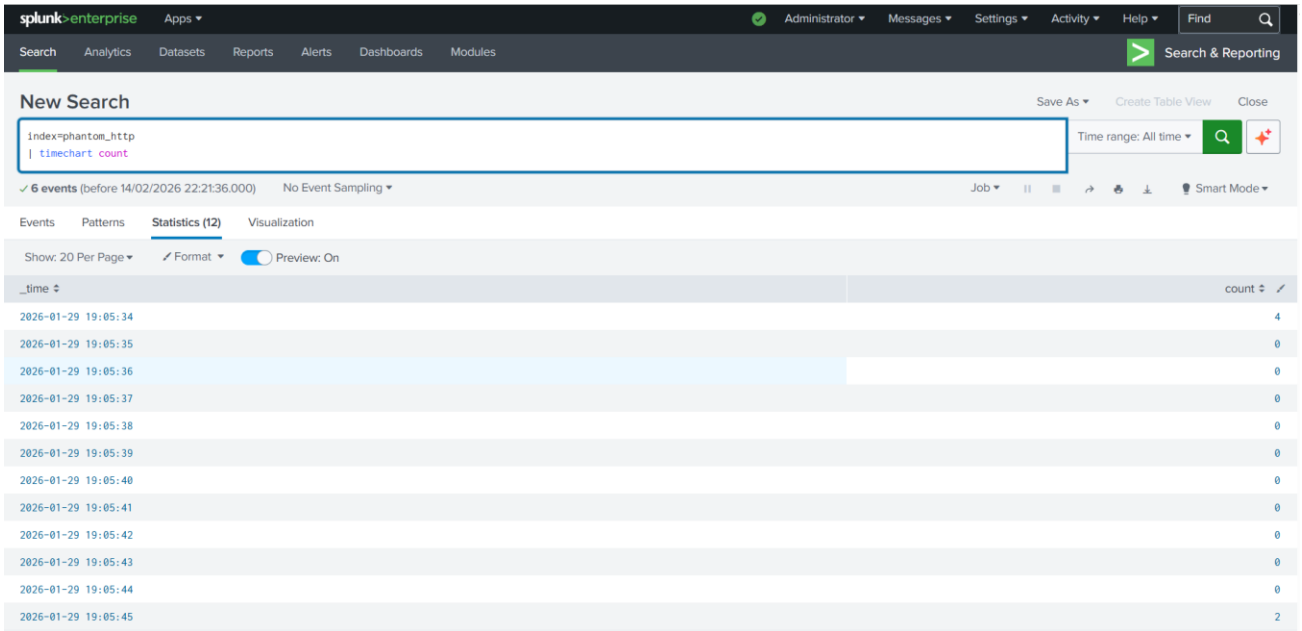


Figure 9: Attack timeline — all 6 transactions within a 40-second burst window on 2026-01-29

```
19:05:34 UTC  → GET /arquivo_20260129190534.txt  [HTTP 200 OK — Config #1]
19:05:45 UTC  → GET /arquivo_20260129190545.txt  [HTTP 200 OK — Config #2]

Total activity window: ~40 seconds — burst consistent with active C2 session
```

Step 8 — Packet Size Analysis

```
index=phantom_http | stats count avg(Length) max(Length) by Source Destination | sort -count
```

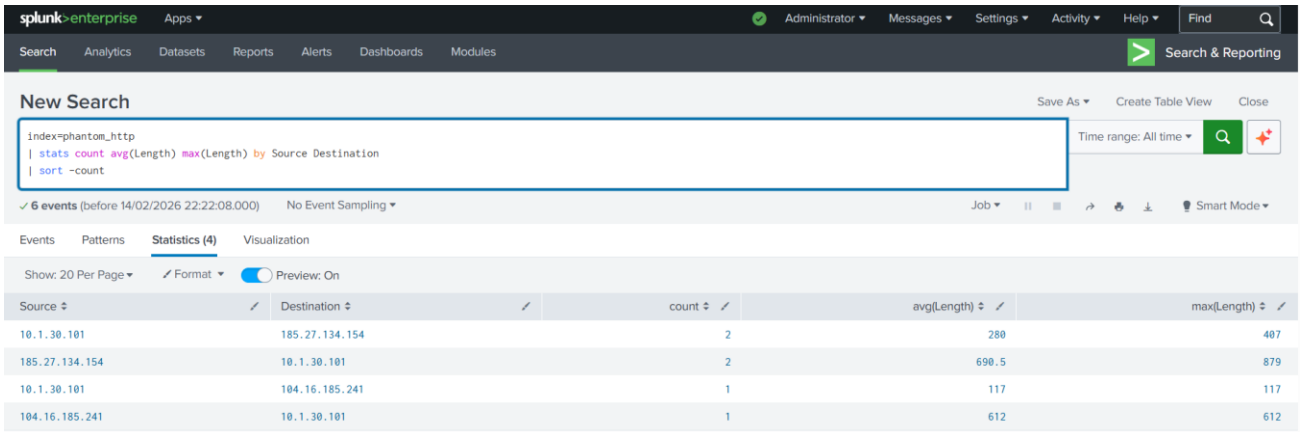


Figure 10: Packet size analysis — larger response packets from C2 confirm data transfer during config downloads

Step 9 — C2 Communication Confirmation

```
index=phantom_http
| eval C2_Server="185.27.134.154 (scxzswx.lovestoblog.com from DNS)"
| where Destination="185.27.134.154"
| table Time Source Destination Info C2_Server
```

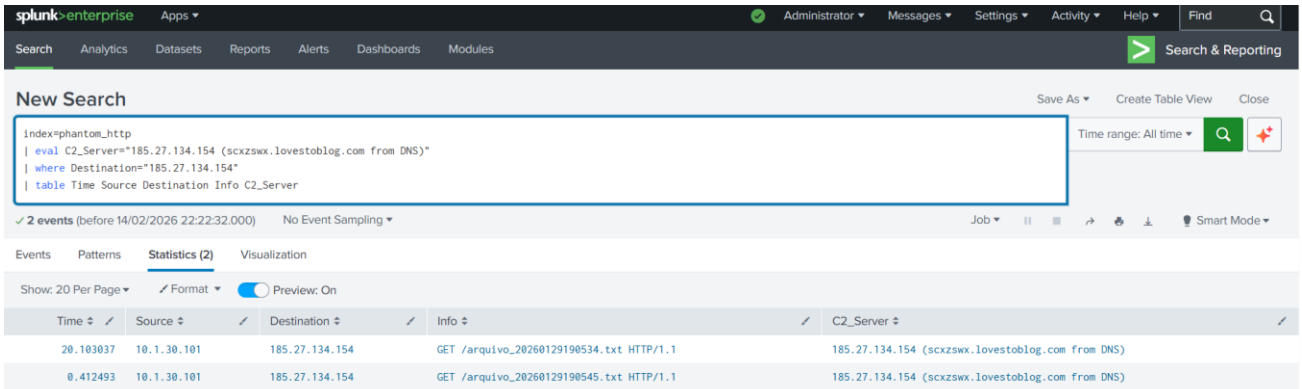


Figure 11: C2 confirmation — HTTP traffic to 185.27.134.154 directly corroborates DNS resolution from Project 1

> **Cross-Protocol Confirmation:** Project 1 identified scxzswx.lovestoblog.com resolving to 185.27.134.154. Project 2 confirms active HTTP communication to that exact IP. Together, these constitute definitive C2 evidence.

Step 10 — Public IP Discovery

```
index=phantom_http
| where Destination="104.16.185.241"
```

```
| eval Service="icanhazip.com (Public IP Discovery)"
| table Time Source Destination Service Info
```

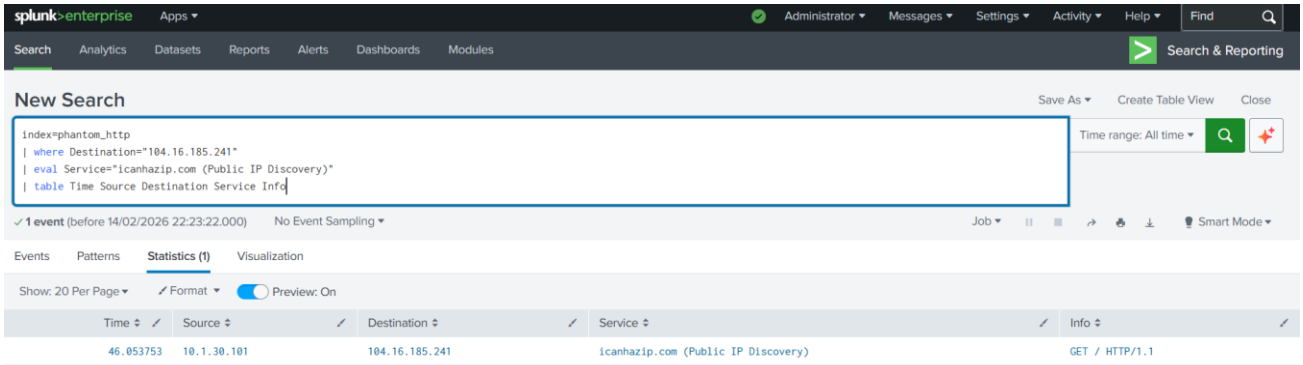


Figure 12: Public IP discovery — HTTP GET to icanhazip.com immediately precedes C2 contact

4.4 Project 2 — Complete Attack Chain (DNS + HTTP)

```
STEP 1 — INFECTION: Host 10.1.30.101 compromised with PhantomStealer

STEP 2 — RECONNAISSANCE (HTTP)
  GET http://icanhazip.com → 104.16.185.241
  Purpose: Discover public IP, identify victim for C2 operator

STEP 3 — C2 RESOLUTION (DNS — Project 1)
  DNS: scxzswx.lovestoblog.com → 185.27.134.154
  DNS: exczx.com → 185.38.151.11

STEP 4 — CONFIG DOWNLOAD (HTTP)
  GET 185.27.134.154/arquivo_20260129190534.txt  [200 OK]
  GET 185.27.134.154/arquivo_20260129190545.txt  [200 OK]
  Malware receives updated C2 instructions

STEP 5 → Encrypted TLS channel established  [Project 3]
```

4.5 MITRE ATT&CK Mapping — Project 2

ID	Technique	Tactic	Evidence
T1016	System Network Configuration Discovery	Discovery	HTTP GET to icanhazip.com — public IP lookup
T1071.001	App Layer Protocol: Web Protocols	Command & Control	HTTP used to contact C2 server 185.27.134.154
T1105	Ingress Tool Transfer	Command & Control	arquivo *.txt config files downloaded via HTTP GET
T1102	Web Service	Command & Control	Standard web protocols abused for covert C2 channel

Table 8: MITRE ATT&CK Mapping — Project 2

5. Project 3 — TLS/SSL Traffic Analysis

Title: *Encrypted C2 Channel Detection using SNI Extraction*

MITRE Techniques: T1573 · T1071.001

5.1 Objective

Project 3 completes the three-layer investigation. Modern malware increasingly uses HTTPS to disguise C2 traffic as legitimate encrypted web browsing, making it invisible to standard packet inspection. This phase demonstrates how encrypted C2 channels can still be detected — without decrypting the traffic — by extracting the Server Name Indication (SNI) field from TLS Client Hello packets, which are transmitted in plaintext before encryption is activated.

> **Why SNI Matters:** The SNI field in a TLS Client Hello contains the target domain name in plaintext. Even in fully encrypted HTTPS sessions, analysts can identify which domain a host is connecting to by examining this field — without any key material or decryption infrastructure.

5.2 Data Acquisition and Ingestion

Applied Wireshark display filter to capture all encrypted traffic:

```
tls || ssl
```

223 TLS events identified. Exported as tls_logs.csv and ingested into phantom_tls.

> **Result:** 223 TLS events successfully indexed — 37x more than HTTP, reflecting the breadth of encrypted communication.

5.3 Threat Hunting Analysis

Step 2 — TLS Traffic Overview

```
index=phantom_tls | table Time Source Destination Info
```

Show: 20 Per Page · Format · Preview: On

< Prev 1 2 3 4 5 6 7 8 ... Next >

Time	Source	Destination	Info
19.433490	104.16.78.6	10.1.30.101	Application Data
19.433451	104.16.78.6	10.1.30.101	Application Data
19.433391	104.16.78.6	10.1.30.101	Application Data, Application Data
19.433276	104.16.78.6	10.1.30.101	Application Data
19.433249	104.16.78.6	10.1.30.101	Application Data
19.433196	104.16.78.6	10.1.30.101	Application Data, Application Data
19.433115	104.16.78.6	10.1.30.101	Application Data
19.433114	104.16.78.6	10.1.30.101	Application Data, Application Data
19.433026	104.16.78.6	10.1.30.101	Application Data, Application Data
19.432036	104.16.78.6	10.1.30.101	Application Data
19.432013	104.16.78.6	10.1.30.101	Application Data
19.349716	10.1.30.101	104.16.78.6	Change Cipher Spec, Application Data, Application Data
19.339588	104.16.78.6	10.1.30.101	Application Data
19.339587	104.16.78.6	10.1.30.101	Server Hello, Change Cipher Spec
19.288031	10.1.30.101	104.16.78.6	Client Hello (SNI=res.cloudinary.com)
19.606133	104.16.78.6	10.1.30.101	Application Data
19.605264	104.16.78.6	10.1.30.101	Application Data
19.600290	104.16.78.6	10.1.30.101	Application Data
19.599998	104.16.78.6	10.1.30.101	Application Data
19.592955	104.16.78.6	10.1.30.101	Application Data

Figure 13: TLS traffic overview — 223 events all originating from infected host 10.1.30.101

Step 3 — TLS Handshake Phase Analysis

Understanding TLS handshake phases is critical before hunting encrypted traffic:

```
index=phantom_tls
| eval phase=case(
    match(Info,"Client Hello"),"Client Hello",
    match(Info,"Server Hello"),"Server Hello",
    match(Info,"Change Cipher Spec"),"Change Cipher Spec",
    match(Info,"Application Data"),"Application Data",
    1=1,"Other")
| stats count by phase
| sort -count
```

TLS Phase	Analyst Notes
Client Hello	Initiates TLS connection. Contains SNI in plaintext — the critical field for domain detection without decryption.
Server Hello	Server responds with chosen cipher suite and certificate. Certificate analysis possible at this stage.
Change Cipher Spec	Signals encryption is now active. All subsequent traffic is fully encrypted and cannot be read without the key.
Application Data	Encrypted payload exchange. C2 commands, exfiltrated data, or config updates occur here. Cannot be inspected without decryption.

Table 9: TLS Handshake Phase Breakdown and Analyst Notes

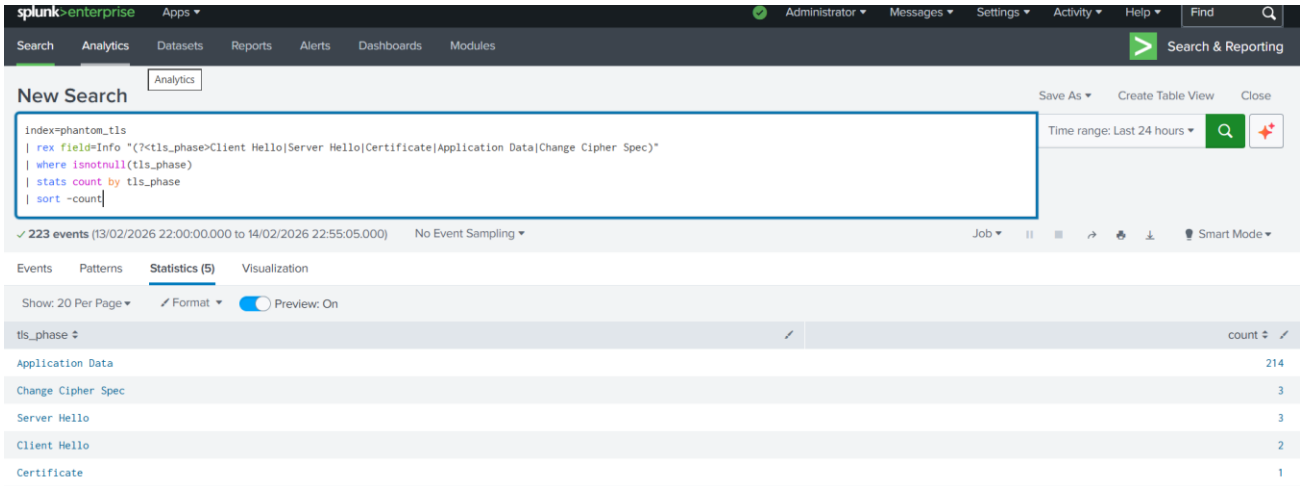


Figure 14: TLS handshake phase distribution — Application Data dominates, confirming active encrypted C2 sessions

Step 4 — Server Name Indication (SNI) Extraction

This is the most critical query of the entire investigation. By targeting Client Hello packets and extracting the SNI field, we identify malicious domains without decryption:

```
index=phantom_tls "Client Hello"
| rex field=Info "SNI=(?<domain>[^\)]+)"
| where isnotnull(domain)
| stats count by domain
| sort -count
```

Domain (SNI)	Status	Evidence / Notes	Severity
exczx.com	**MALICIOUS C2**	Confirmed via DNS (Project 1) + HTTP (Project 2) + TLS SNI (Project 3)	**CRITICAL**
res.cloudinary.com	**UNDER REVIEW**	Legitimate CDN — common malware staging technique; further correlation recommended	**SUSPICIOUS**

Table 10: Domains Extracted via SNI from TLS Client Hello Packets

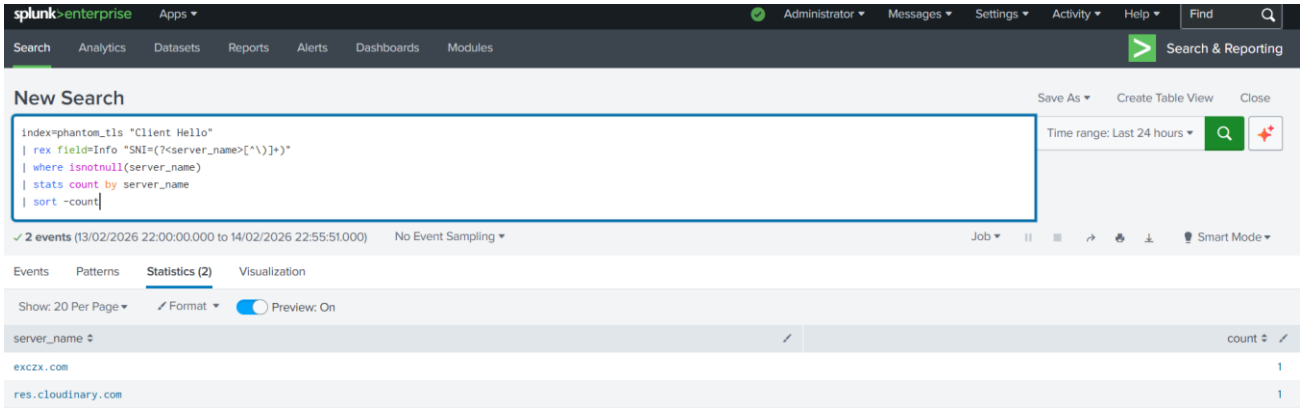


Figure 15: SNI extraction result — exczx.com confirmed as malicious C2 across all three protocol layers

> **Critical Finding — Triple-Layer Confirmation:** exczx.com was detected in DNS (Project 1: 2 queries), confirmed via HTTP (Project 2: traffic to resolved IP), and now confirmed via TLS SNI (Project 3). This triple-layer confirmation constitutes unambiguous C2 infrastructure evidence.

Step 5 — Source and Destination Analysis

```
index=phantom_tls | stats count by Source Destination | sort -count
```

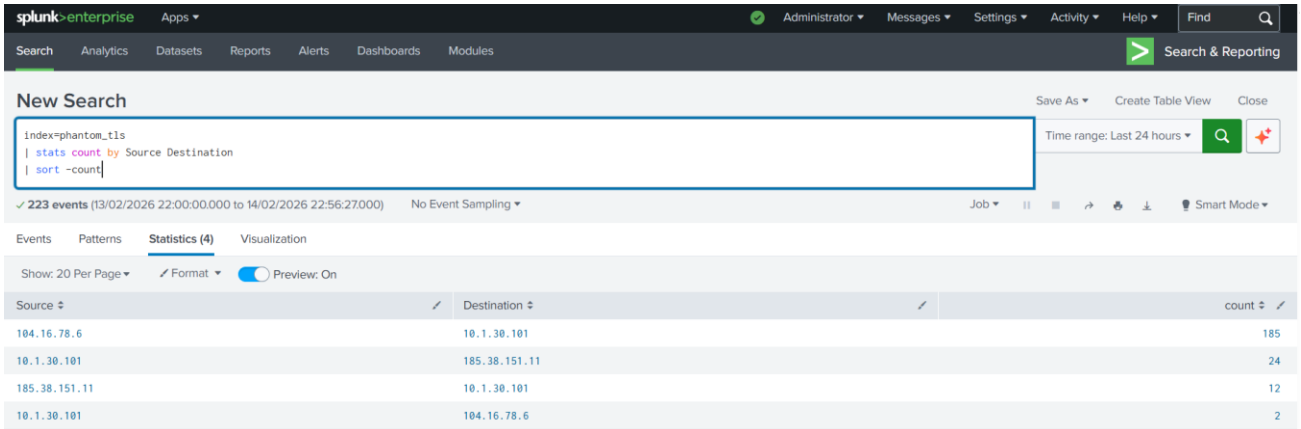


Figure 16: TLS source-destination analysis — multiple encrypted endpoints contacted by infected host

Step 6 — Malicious Domain TLS Session

```
index=phantom_tls
| search Info="*exczx.com*"
| table Time Source Destination Info
```

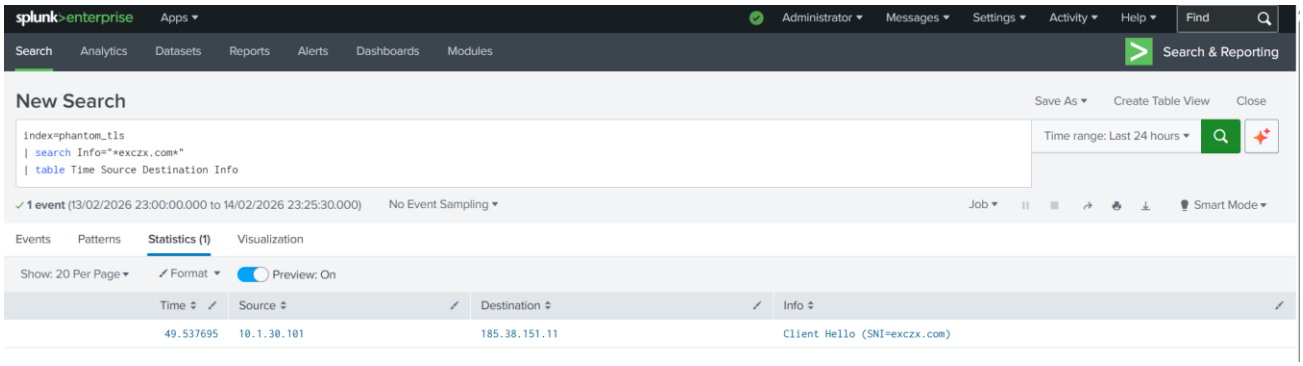


Figure 17: Malicious domain TLS session — full handshake from Client Hello through Application Data for exczx.com

The captured TLS session for exczx.com progresses through all handshake phases before entering Application Data. The payload content cannot be read without the private key, but the confirmed domain, session establishment, and substantial Application Data volume provide definitive evidence of an active encrypted C2 channel.

Step 7 — CDN Traffic Analysis

```
index=phantom_tls
| search Info="*cloudinary*"
| table Time Source Destination Info
```

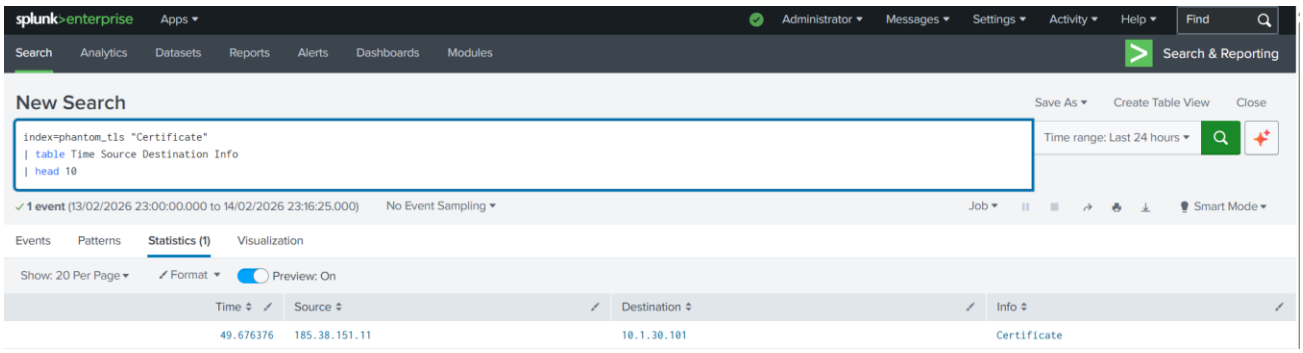


Figure 18: CDN and certificate analysis — res.cloudinary.com TLS connections identified; correlation with HTTP logs recommended

> **Recommendation:** Correlate Cloudinary TLS events with HTTP logs from Project 2 to determine if any file downloads from Cloudinary occurred. CDN-hosted payloads are a common evasion technique that blends malware delivery with legitimate traffic.

Step 8 — Encrypted Traffic Volume Analysis

```
index=phantom_tls
| stats sum(Length) as total_bytes count by Source Destination
| eval total_kb=round(total_bytes/1024,2)
| table Source Destination count total_kb
| sort -total_kb
```

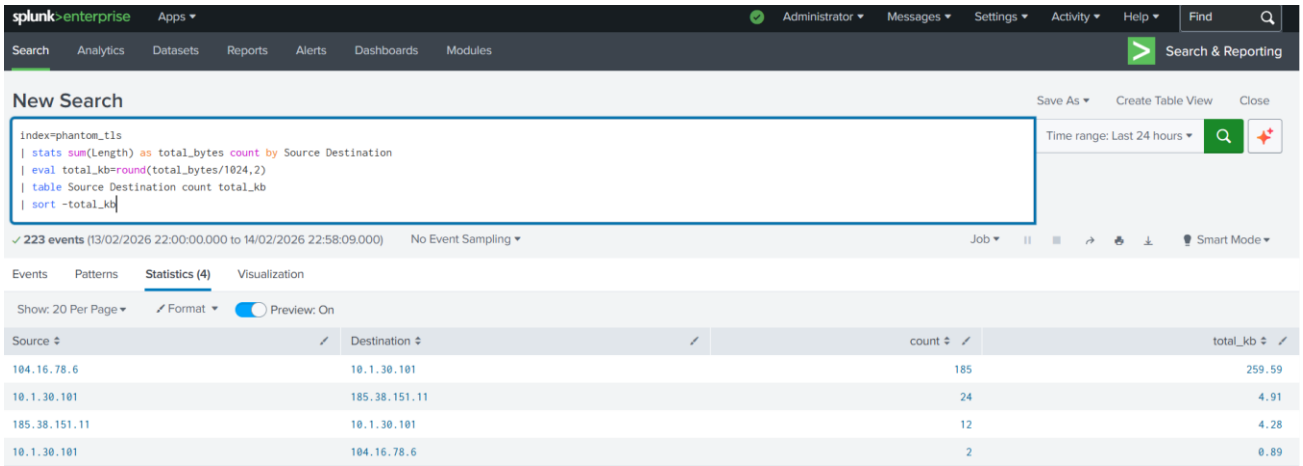


Figure 19: Encrypted data volume — total bytes per destination quantified to identify highest-priority channels

Step 9 — TLS Activity Timeline

```
index=phantom_tls | timechart count
```

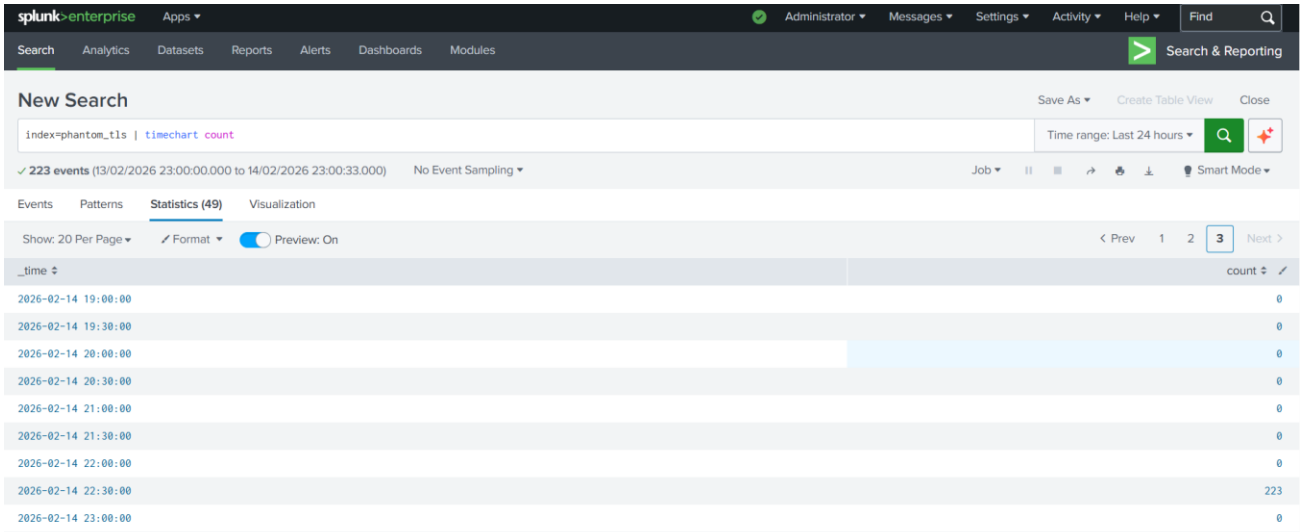


Figure 20: TLS timeline — burst pattern confirming active C2 session window correlating with HTTP activity from Project 2

Step 10 — Complete Attack Chain Confirmation Query

```
index=phantom_tls "Client Hello"
| rex field=Info "SNI=(?<domain>[^\)]+)"
| where domain="exczx.com"
| eval Project_1_DNS="DNS queries to exczx.com detected (2 queries)"
| eval Project_2_HTTP="HTTP traffic to 185.38.151.11 (exczx.com IP)"
| eval Project_3_TLS="TLS encrypted connection confirmed via SNI"
| table Time Source Destination domain
        Project_1_DNS Project_2_HTTP Project_3_TLS
```

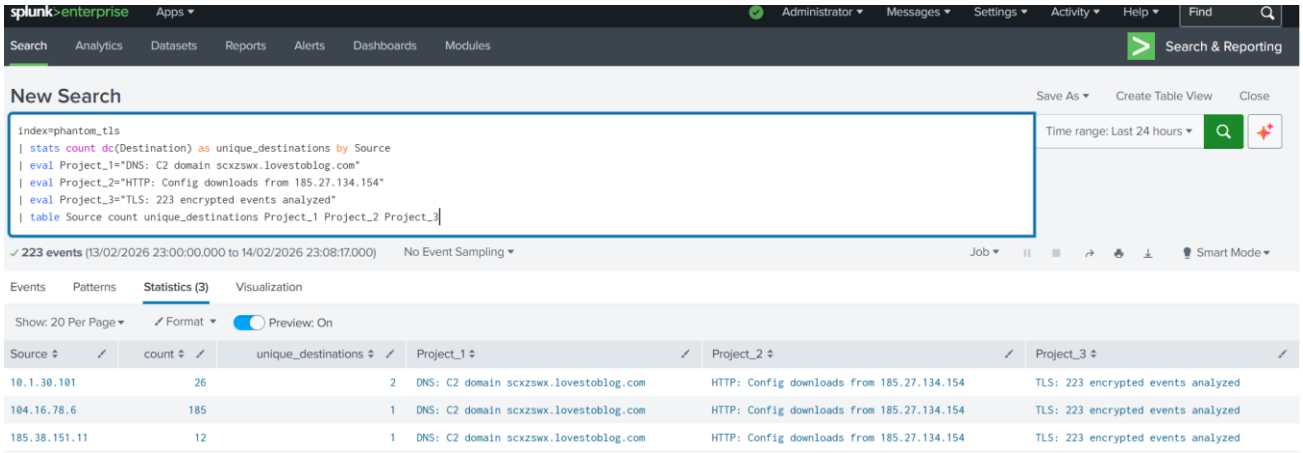


Figure 21: Complete attack chain query — confirms exczx.com as C2 across DNS, HTTP, and TLS in a single result

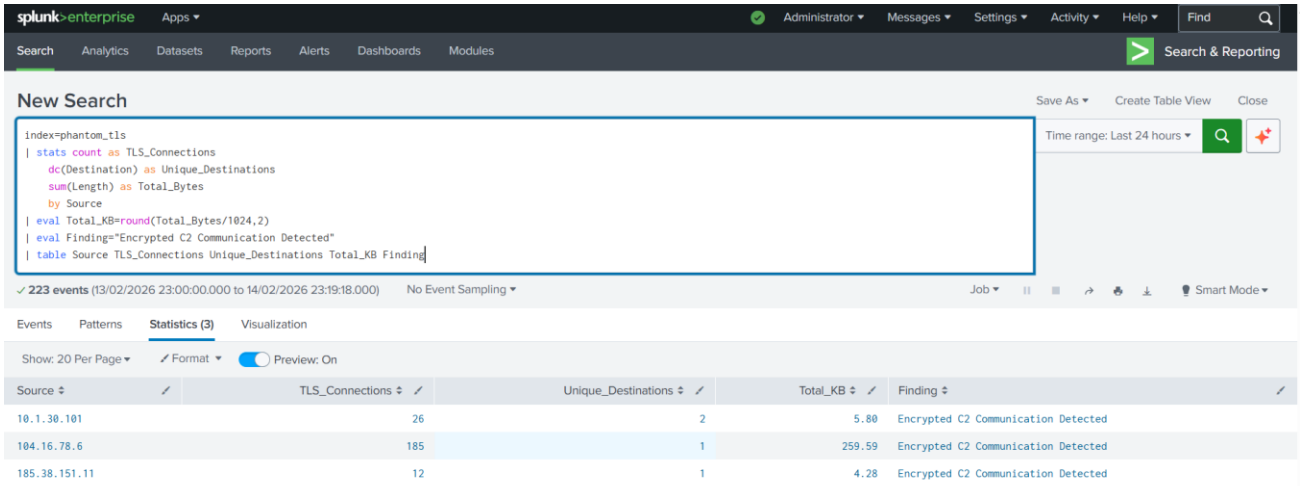


Figure 22: TLS investigation summary — full protocol breakdown and encrypted channel confirmation

5.4 MITRE ATT&CK Mapping — Project 3

ID	Technique	Tactic	Evidence
T1573	Encrypted Channel	Command & Control	TLS HTTPS connection to exczx.com confirmed via SNI extraction
T1071.001	App Layer Protocol: Web	Command & Control	HTTPS used as C2 transport — 223 encrypted events analysed

Table 11: MITRE ATT&CK Mapping — Project 3

6. Complete Attack Chain Reconstruction

6.1 Multi-Protocol Correlation

The three projects collectively reconstruct the complete PhantomStealer infection lifecycle. The table below consolidates every finding from all three projects into a unified view, demonstrating how each protocol layer contributed distinct evidence that — when combined — produces a definitive picture of the C2 infrastructure.

Protocol	Project	Finding	Significance / MITRE
DNS	Project 1	scxzswx.lovestoblog.com queried 4 times	Beaconing — repeated C2 callback (T1071.004)
DNS	Project 1	exczx.com queried 2 times	Secondary C2 resolution confirmed
DNS	Project 1	icanhazip.com queried for public IP	Public IP reconnaissance (T1016)
HTTP	Project 2	HTTP GET to 185.27.134.154	C2 IP confirmed from DNS resolution
HTTP	Project 2	/arquivo_20260129190534.txt downloaded (200 OK)	Config retrieved via HTTP GET (T1105)
HTTP	Project 2	/arquivo_20260129190545.txt downloaded (200 OK)	Updated config retrieved (T1105)
TLS	Project 3	Client Hello with SNI=exczx.com detected	Encrypted C2 channel opened (T1573)
TLS	Project 3	223 TLS events — high Application Data volume	Active encrypted C2 session confirmed

Table 12: Cross-Protocol Correlation — All Findings Across Projects 1, 2, and 3

6.2 PhantomStealer C2 Architecture

The investigation reveals that PhantomStealer employs a dual-channel C2 architecture using both unencrypted HTTP and encrypted HTTPS across two separate C2 domains. This provides operational redundancy: if one channel is detected and blocked, the other continues operating. HTTPS also provides stealth by blending C2 traffic with normal web browsing.

```
PHANTOMSTEALER C2 ARCHITECTURE — 10.1.30.101

 |-- [DNS] --> scxzswx.lovestoblog.com --> 185.27.134.154
 |             (4 queries — primary C2 beaconing)
 |
 |-- [HTTP] --> 185.27.134.154
 |             GET /arquivo_20260129190534.txt  [200 OK]
 |             GET /arquivo_20260129190545.txt  [200 OK]
 |
 |-- [DNS] --> exczx.com --> 185.38.151.11
 |             (2 queries — secondary C2)
 |
 |-- [TLS] --> exczx.com (SNI confirmed)
 |             223 encrypted events
 |             Application Data exchanged
 |
 `-- [HTTP/DNS] --> icanhazip.com --> 104.16.185.241
                    Public IP reconnaissance
```

6.3 Complete Infection Timeline

```
PHASE 1 — INITIAL COMPROMISE
  Host 10.1.30.101 infected with PhantomStealer (execution method outside PCAP scope)

PHASE 2 — RECONNAISSANCE  (~2026-01-29 19:05 UTC)
  DNS query: icanhazip.com (public IP lookup)
  HTTP GET: http://icanhazip.com → victim public IP obtained
  MITRE: T1016

PHASE 3 — C2 DOMAIN RESOLUTION
```

```
  DNS query x4: scxzswx.lovestoblog.com → 185.27.134.154
  DNS query x2: exczx.com → 185.38.151.11
  MITRE: T1071.004

PHASE 4 — UNENCRYPTED C2 CONTACT  (19:05:34–19:05:45 UTC)
  HTTP GET 185.27.134.154/arquivo_20260129190534.txt [200 OK]
  HTTP GET 185.27.134.154/arquivo_20260129190545.txt [200 OK]
  Malware receives updated instructions
  MITRE: T1071.001, T1105

PHASE 5 — ENCRYPTED C2 CHANNEL
  TLS Client Hello → SNI=exczx.com
  Full TLS handshake → encryption active
  Application Data exchanged — 223 TLS events total
  MITRE: T1573, T1071.001
```

7. Full MITRE ATT&CK Coverage

The table below consolidates the complete MITRE ATT&CK coverage across all three projects, spanning two tactics and providing a comprehensive behavioural profile of the PhantomStealer infection.

ID	Technique	Tactic	Evidence	Project
T1016	**System Network Configuration Discovery**	Discovery	Public IP lookup via icanhazip.com	1, 2
T1071.004	**App Layer Protocol: DNS**	Command & Control	4 repeated DNS queries to C2 domain — beaconing confirmed	1
T1071.001	**App Layer Protocol: Web Protocols**	Command & Control	HTTP and HTTPS used to contact C2 servers	2, 3
T1105	**Ingress Tool Transfer**	Command & Control	arquivo_*.txt config files downloaded via HTTP GET	2
T1102	**Web Service**	Command & Control	Standard web protocols abused for covert C2 channel	2
T1573	**Encrypted Channel**	Command & Control	TLS HTTPS confirmed via SNI field in Client Hello packet	3

Table 13: Complete MITRE ATT&CK Coverage — PhantomStealer Investigation (All Projects)

8. Master Indicators of Compromise

All IOCs extracted across the three-project investigation are consolidated below. These should be immediately operationalised by feeding into firewall blocklists, DNS sinkholes, SIEM detection rules, and threat intelligence platforms.

IOC Type	Value	Found In	Severity
C2 Domain	scxzswx.lovestoblog.com	Projects 1, 2	**CRITICAL**
C2 Domain	exczx.com	Projects 1, 3	**CRITICAL**

IOC Type	Value	Found In	Severity
C2 IP Address	185.27.134.154	Project 2	**CRITICAL**
C2 IP Address	185.38.151.11	Projects 1, 3	**CRITICAL**
Recon IP	104.16.185.241 (icanhazip.com)	Projects 1, 2	**HIGH**
Infected Host	10.1.30.101	All Projects	**CRITICAL**
Network Segment	10.1.30.0/24	Project 1	**INFO**
Downloaded File	/arquivo_20260129190534.txt	Project 2	**HIGH**
Downloaded File	/arquivo_20260129190545.txt	Project 2	**HIGH**
TLS SNI	exczx.com (Client Hello)	Project 3	**CRITICAL**

Table 14: Master IOC List — All Projects Combined

9. Recommendations

9.1 Prioritised Response Actions

Priority	Action	Detail
CRITICAL	**Isolate 10.1.30.101**	Disconnect from all networks immediately; preserve RAM and disk image for forensic analysis
CRITICAL	**Block C2 IPs at Firewall**	Drop all traffic to/from 185.27.134.154 and 185.38.151.11
CRITICAL	**DNS Sinkhole C2 Domains**	Redirect scxzswx.lovestoblog.com and exczx.com to sinkhole server
HIGH	**Deploy TLS/SSL Inspection**	Implement SSL inspection proxy to decrypt and inspect HTTPS C2 channels
HIGH	**Enable SNI-Based Filtering**	Configure firewall to block TLS connections to known-bad domains via SNI
HIGH	**Block IP Lookup Services**	Prevent internal hosts accessing icanhazip.com, ipinfo.io, api.ipify.org
MEDIUM	**Deploy EDR on All Endpoints**	Endpoint Detection and Response for process, file, and network telemetry
MEDIUM	**Enable Full DNS Logging**	Log all DNS queries from all endpoints; feed into Splunk for correlation
LOW	**Threat Intelligence Feeds**	Subscribe to PhantomStealer IOC feeds; auto-enrich SIEM with new indicators

Table 15: Prioritised Incident Response Recommendations

9.2 Production SIEM Detection Rules

Rule 1 — DNS Beaconing Detection

Alert when any host queries the same domain more than 3 times in 1 hour:

```
index=dns_logs earliest=-1h
| rex field=query "(?<domain>[a-zA-Z0-9\.-]+\.[a-zA-Z]{2,})"
| stats count by src_ip domain
| where count > 3
| eval alert_type="DNS_BEACONING_DETECTED"
```

Rule 2 — Public IP Discovery Detection

```
index=dns_logs OR index=http_logs
| where dest_domain IN ("icanhazip.com","ipinfo.io","api.ipify.org","checkip.amazonaws.com")
| stats count by src_ip dest_domain
| eval alert_type="PUBLIC_IP_LOOKUP"
```

Rule 3 — HTTP Config File Download Pattern

```
index=http_logs
| rex field=url "(?<pattern>\/arquivo_\d{14}\.txt)"
| where isnotnull(pattern)
| eval alert_type="PHANTOMSTEALER_CONFIG_DOWNLOAD"
```

Rule 4 — SNI-Based C2 Detection

```
index=tls_logs "Client Hello"
| rex field=sni "(?<domain>[a-zA-Z0-9\.-]+)"
| where domain IN ("exczx.com","scxzswx.lovestoblog.com")
| stats count by src_ip domain
| eval alert_type="SNI_C2_DOMAIN_DETECTED"
```

Rule 5 — High-Volume Encrypted Traffic Anomaly

```
index=tls_logs "Application Data"
| stats sum(bytes) as total_bytes by src_ip dest_ip
| where total_bytes > 100000
| eval alert_type="HIGH_VOLUME_ENCRYPTED_TRAFFIC"
```

9.3 Long-Term Security Improvements

- **Deploy SSL/TLS Inspection Gateway — decrypt and inspect HTTPS from internal hosts to unknown external domains; this would allow payload inspection of the encrypted C2 channel from Project 3.**
- **Implement DNS Security Filtering — block queries to malicious domains at resolution time, preventing C2 contact before any TCP connection is established.**
- **Enable Comprehensive Network Logging — collect DNS, HTTP, and TLS logs from all endpoints; centralise in Splunk with long-term retention for retrospective threat hunting.**
- **Deploy EDR on All Endpoints — provide host-level telemetry (process execution, file writes, registry) to complement network-level analysis.**
- **Establish Traffic Baselines — define normal DNS query volumes, HTTP patterns, and encrypted traffic volumes per host; automate alerting on deviations.**
- **Integrate Threat Intelligence Feeds — subscribe to PhantomStealer IOC feeds; configure automatic SIEM enrichment with current indicators.**

10. Conclusion

This investigation successfully reconstructed the complete Command and Control infrastructure of a PhantomStealer malware infection across three distinct network protocol layers. By analysing DNS, HTTP, and TLS traffic independently and then correlating findings across all three, I was able to build a definitive picture of how this malware communicates with its operators and what actions it takes against infected hosts.

The key achievement is the triple-layer confirmation of the exczx.com C2 domain — first detected through DNS query analysis, confirmed through HTTP session analysis, and verified through TLS SNI extraction. This multi-layer approach is the gold standard for C2 infrastructure identification, as it eliminates false positives and provides irrefutable evidence of malicious activity.

Equally important is the SNI extraction technique demonstrated in Project 3. By examining the TLS Client Hello before encryption is activated, analysts can identify malicious HTTPS destinations without decryption infrastructure — a critical capability in environments where the majority of traffic is now encrypted.

The investigation also revealed PhantomStealer's dual-channel C2 design: both unencrypted HTTP (for fast config retrieval) and encrypted HTTPS (for stealth and command execution), across two separate domains on different IP ranges. This redundancy makes the malware resilient against single-domain blocking and single-IP takedowns.

> **Key Takeaway:** Even with a small sample (239 events across three protocol layers), structured multi-protocol analysis reveals the complete operational behaviour of sophisticated malware. The DNS-to-HTTP-to-TLS correlation methodology demonstrated here is directly applicable to real-world SOC investigation workflows.

10.1 Skills Demonstrated

Skill Area	Demonstration
Threat Hunting Methodology	Systematic protocol-by-protocol analysis from broad data to specific IOCs
SIEM Query Development (SPL)	20+ custom SPL queries including regex extraction, stats, timechart, and correlation
DNS Traffic Analysis	Beaconing detection, domain frequency analysis, compromised host identification
HTTP Protocol Analysis	C2 session identification, payload detection, attack timeline reconstruction
Encrypted Traffic Analysis	TLS handshake analysis and SNI extraction for domain detection without decryption
Cross-Protocol Correlation	Linked DNS findings to HTTP evidence to TLS confirmation across three projects
IOC Extraction	10 network, host, file, and behavioural IOCs extracted and documented
MITRE ATT&CK Mapping	6 techniques mapped across Discovery and Command & Control tactics
Professional Documentation	Structured investigation report with tables, timelines, code blocks, and recommendations

Table 16: Skills Demonstrated Across the Three-Project Investigation

11. References

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Report prepared by Rohit Aswal · February 2026

SOC Analyst · Threat Hunter · Certified Ethical Hacker (CEH)

GitHub: github.com/rohitaswal2108-Roh · LinkedIn: linkedin.com/in/rohit-aswal08

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